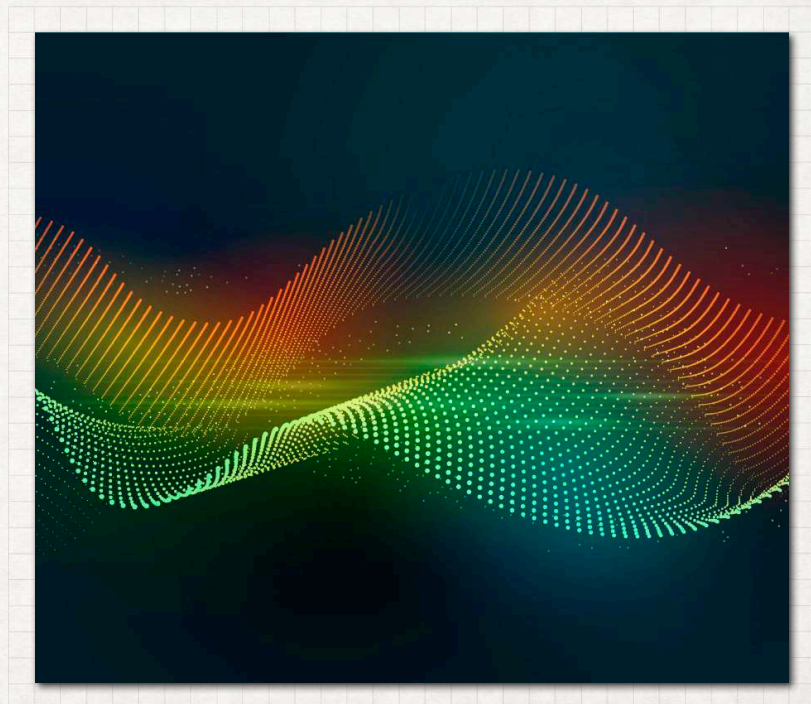
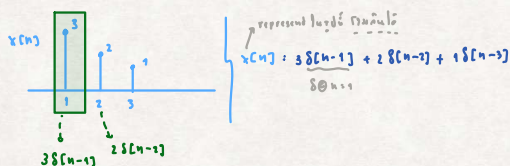


Note Lecture 1 min!

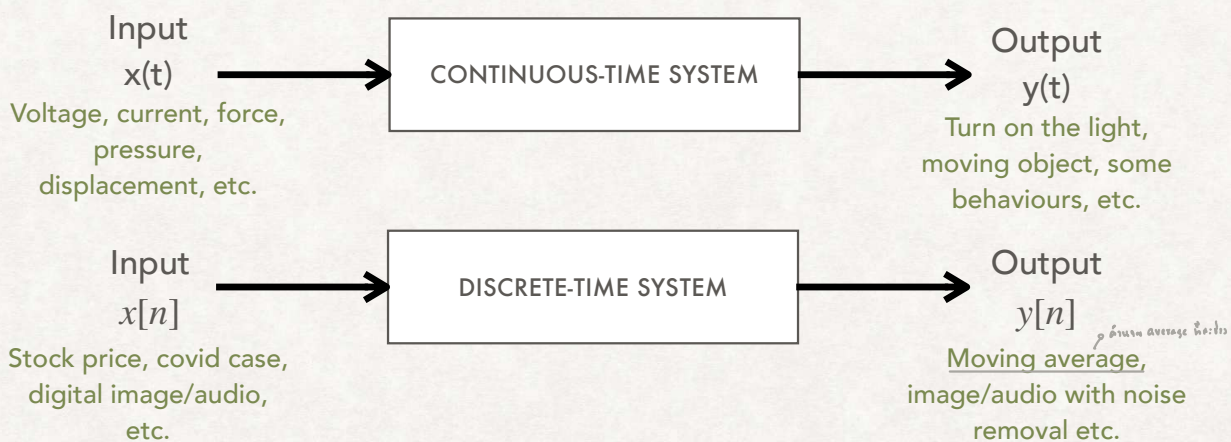
$x[n]$, $x(t)$, $T: \omega_1: 1T_2 \rightarrow \frac{1}{T}$ mod 2π : periodic
 $\delta[n]$, $\delta(t)$ $\xrightarrow{\text{Fourier}}$ $X(\omega) \delta(t - t_0) = X(\omega) \cdot \delta(t - t_0)$ $\xrightarrow{\text{Const.}}$ $\frac{\delta(\omega)}{\delta(t)}$
 Sifting: $\int_{-\infty}^{\infty} x(t) \delta(t - t_0) dt = \begin{cases} x(t_0) & ; t_0 \in \mathbb{R} \\ 0 & ; \text{otherwise} \end{cases}$ $\delta(t \rightarrow 0) = \infty$
 $\delta[n \rightarrow 0] = 1$
 Scale: $\delta(at + b) = \frac{1}{|a|} \delta(t + \frac{b}{a})$

LECTURE 02 CONVOLUTION



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CONTINUOUS & DISCRETE SYSTEMS



AKA Operation from Input $\rightarrow$ Output
Convolutions - Analysis of linear system can be reduced to the study of the system to basic input signals

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CLASSIFICATION OF SYSTEMS

System is the process for transforming input resulting in a desired output. There are a number of basic properties of CT and DT systems.

- Linear / Non-linear
- Time-varying, Time-invariant
- System with/without memory
- Causal system (non-anticipatory)
- Invertibility / Inverse system
- Stable systems (BIBO system)

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Basic system properties

1 Linearity



The system is linear if:

- (1) The response to $x_1(t) + x_2(t)$ is $y_1(t) + y_2(t)$: additivity properties
- (2) The response to $ax_1(t)$ is $ay_1(t)$, where a is any complex instant : scaling properties

Two properties defining a linear system can be combined into a single statement:

$$\text{continuous-time: } ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t)$$

$$\text{discrete-time: } ax_1[n] + bx_2[n] \rightarrow ay_1[n] + by_2[n]$$

a & b are complex constants

If $x_k[n]$, $k = 1, 2, 3, \dots$ are sets of inputs of a discrete time linear system with the corresponding outputs $y_k[n]$, $k = 1, 2, 3, \dots$, then the response of these inputs given by

$$x[n] = \sum_k a_k x_k[n] = a_1 x_1[n] + a_2 x_2[n] + a_3 x_3[n] + \dots$$

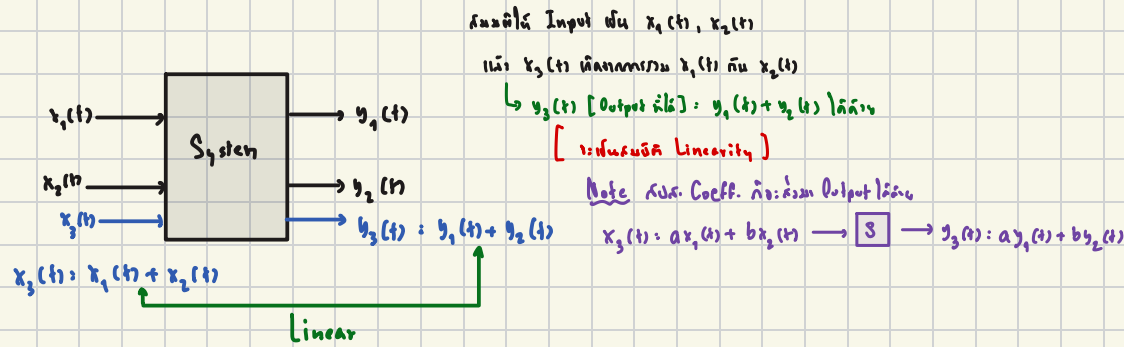
is

$$y[n] = \sum_k a_k y_k[n] = a_1 y_1[n] + a_2 y_2[n] + a_3 y_3[n] + \dots$$

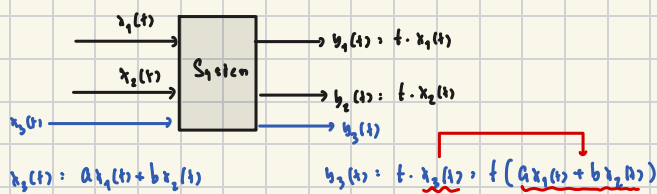
} superposition properties for both continuous & discrete time.

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Linearity



Q: $y(t) = t \cdot x(t) \rightarrow$ is it Linear?



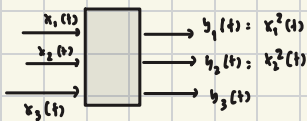
Check Linear?

LHS: $y_3(t) = t \cdot x_3(t)$

RHS: $y_3(t) = t \cdot (ax_1(t) + bx_2(t)) = a \cdot t \cdot x_1(t) + b \cdot t \cdot x_2(t)$
 $= ay_1(t) + by_2(t)$

$\therefore$ is Linear (same output)

Q2: $y(t) = x^2(t)$



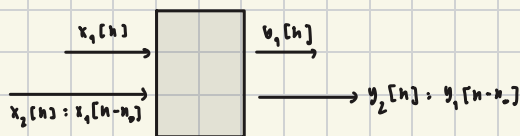
Check Linear?

LHS: $y_3(t) = x_3^2(t) = a^2 x_1^2(t) + b^2 x_2^2(t)$

RHS: $y_3(t) = a^2 x_1^2(t) + 2 \cdot ax_1(t) \cdot bx_2(t) + b^2 x_2^2(t) \neq$
 $= a^2 y_1(t) + b^2 y_2(t) + 2 \cdot ax_1(t) \cdot bx_2(t)$

$\therefore$ Non-Linear

Time-Variant

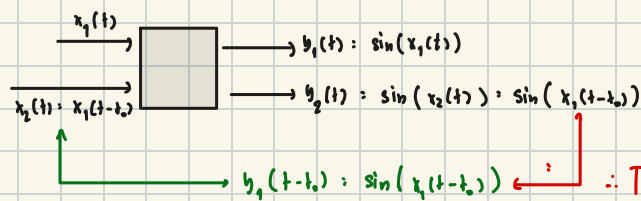


Example I/O of $x_1[n]$ / $x_2[n]$ with $y_1[n]$ / $y_2[n]$

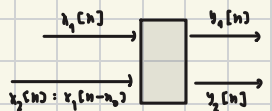
if $n_2[n] = x_2[n - n_0] \Rightarrow$ delay

if $y_2[n] = y_2[n - n_0]$ (delay) $\Rightarrow$ is Time-Invariant (delay)

Ex1: $y(t) = \sin(x(t))$



Ex2: $y[n] = n \cdot x[n]$



$y_2[n] = n \cdot x_2[n] = n \cdot x_1[n - n_0]$

if $y_1[n]$ and $x_1[n]$ are the same $y_2[n]$ also?

Check time-variant
 LHS: $y_1[n - n_0] = (n - n_0) x_1[n - n_0]$
 RHS: $y_2[n] = n \cdot x_1[n - n_0] \neq$

$\therefore$ is Time-Variant $\sim$ Output varies with Shift

Ex. System S

$$y(t) = tx(t)$$

[linear] $x_1(t) \rightarrow y_1(t)$

$$y_1(t) = tx_1(t)$$

$$y_2(t) = tx_2(t)$$

$$x_3(t) = ax_1(t) + bx_2(t)$$

$$y_3(t) = atx_1(t) + btx_2(t)$$

$$= ay_1(t) + by_2(t)$$

Ex. $y(t) = x^2(t)$

[non-linear]

$$x_1(t) \rightarrow y_1(t) = x_1^2(t)$$

$$y_2(t) = x_2^2(t)$$

$$ax_1(t) + bx_2(t) \rightarrow y_3(t)$$

$$y_3(t) = (ax_1(t) + bx_2(t))^2$$

$$(1 \cdot 2 + 0)^2 = 4$$

$$\text{e.g. } x_1(t) = 1$$

to be linear system:

$$y_3(t) = ay_1(t) + by_2(t)$$

$$y_3(t) = ax_1^2(t) + bx_2^2(t)$$

not equivalent

$$1^2 \cdot 2 + 0 = 2$$

$$x_2(t) = 2, a = 2, b = 0$$

2. Time-invariance

A system is time-invariant if the behavior and characteristics are fixed over time.

A system is time-invariant if a time shift in the input signal results in an identical time shift in the output signal.

$$x[n] \rightarrow y[n]$$

$$x[n-n_0] \rightarrow y[n-n_0]$$

$$x(t) \rightarrow y(t)$$

$$x(t-t_0) \rightarrow y(t-t_0)$$

Ex $y[n] = n x[n]$ [time-varying]

$$\text{if } x_1[n] = \delta[n]$$

$$y_1[n] = n \delta[n] = 0$$

$$\text{if } x_2[n] = \delta[n-1]$$

$$y_2[n] = n \delta[n-1] = 1 \cdot \delta[n-1]$$

$x_2[n]$ is a shifted version of $x_1[n]$

but $y_2[n]$ is not a shifted version

of $y_1[n]$

$\therefore$ time-varying system

Ex $y(t) = \sin(x(t))$ [time-invariance]

$$y_1(t) = \sin(x_1(t))$$

$$y_2(t) = \sin(x_2(t))$$

$$\text{if } x_2(t) = x_1(t-t_0)$$

$$y_2(t) = \sin(x_1(t-t_0))$$

$$y_1(t-t_0) = \sin(x_1(t-t_0))$$

also

3. system with/without memory

A system is memoryless if its output of each value of the independent variable at a given time is dependent only on the input at the same time.

Ex. $y[n] = (2x[n] - x^2[n])^2$ is memoryless $\Rightarrow$ $x(t), x[n]$ $\Rightarrow$ present time only

$$y[n] = \sum_{k=-\infty}^n x[k] \quad // \text{accumulator}$$

is a system w/ memory $\Rightarrow$ $\text{present time + past}$

Ex. $y[n] = x[n+2] \Rightarrow$ $\text{future} = \text{Memory}$

4. causal system (non-anticipatory)

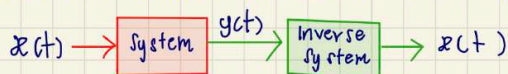
A system is causal if output at any time t_0 depends only on values of input @ $t \leq t_0$

Ex. $y[n] = x[n] + x[n+1]$ $\leftarrow$ advanced time $\Rightarrow$ Non-Causal $(\because \text{future})$

All memoryless systems are causal

5. Invertibility (Inverse System)

A system is said to be invertible if distinct inputs leads to distinct outputs.



Ex. $y(t) = 2x(t) \rightarrow$ inverse $z(t) = \frac{1}{2}y(t)$

$y(t) = \cos(t)$ $\rightarrow$ non-invertible
 $x(t)$ repeat at $x(t) + 2\pi$

6. Stable system (BIBO systems) "stability"

Signal $x(t)$ is said to be bounded if its magnitude doesn't grow without bound

$$|x(t)| < B < \infty \quad \text{for all } t$$

A system is BIBO stable (bounded-input bounded output) for any bounded input $x(t)$, the response $y(t)$ is also bounded.
 $|x(t)| < B_1 < \infty$ implies $|y(t)| < B_2 < \infty$

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Output on Bound $\Rightarrow$ BIBO

Ex. $y(t) = e^{x(t)}$ if $|x(t)| < B$, then $|y(t)| = |e^{x(t)}| < e^B < \infty$

$$y(t) = \int_{-\infty}^t x(\tau) d\tau \quad \text{If } x(t) = u(t), y(t) = \int_{-\infty}^t u(\tau) d\tau = r(t)$$

if $|u(t)| < B$, $r(t)$ is unbounded, system is not stable.
 $\text{Bound Input} \Rightarrow \text{Output is Bound}$

References: Section 1.6, A.V. Oppenheim & A.S. Willsky

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LINEAR-TIME INVARIANT SYSTEMS (LTI)

Output of the system is given by

(1) **superposition property (Linearity)** - the response of the input is given by

$$x[n] = \sum_k a_k x_k[n] = a_1 x_1[n] + a_2 x_2[n] + a_3 x_3[n] + \dots$$

is

$$y[n] = \sum_k a_k y_k[n] = a_1 y_1[n] + a_2 y_2[n] + a_3 y_3[n] + \dots$$

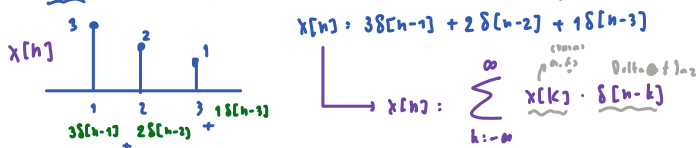
(2) **Time - invariance** - the system is time-invariance if a time shift in the input signal results in an identical time shift in the output signal

$$\text{If } x[n] \rightarrow y[n]$$

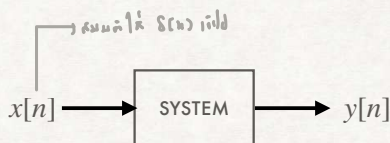
For time-invariance system, we get $x[n - n_0] \rightarrow y[n - n_0]$

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Note Discrete Signal can be decomposed into $\delta[n]$ signals:



DISCRETE-TIME SIGNAL CONVOLUTION



Given $h[n]$ is the response of a linear system to the input $\delta[n]$.



For the discrete-time system, input $x[n]$ can be written as the weighted sum of shifted unite-sampled functions:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

If the system is **time-invariance**, $\delta[n-k] \rightarrow h[n-k]$

From superposition properties, (Linearity)

$$x[n] \rightarrow y[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \rightarrow y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] = \sum_{k=-\infty}^{\infty} x[n-k] h[k] = x[n] * h[n]$$

"CONVOLUTION"

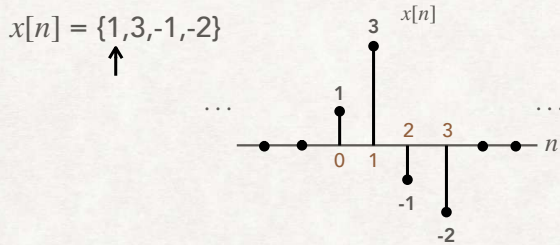
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Note From Convolution

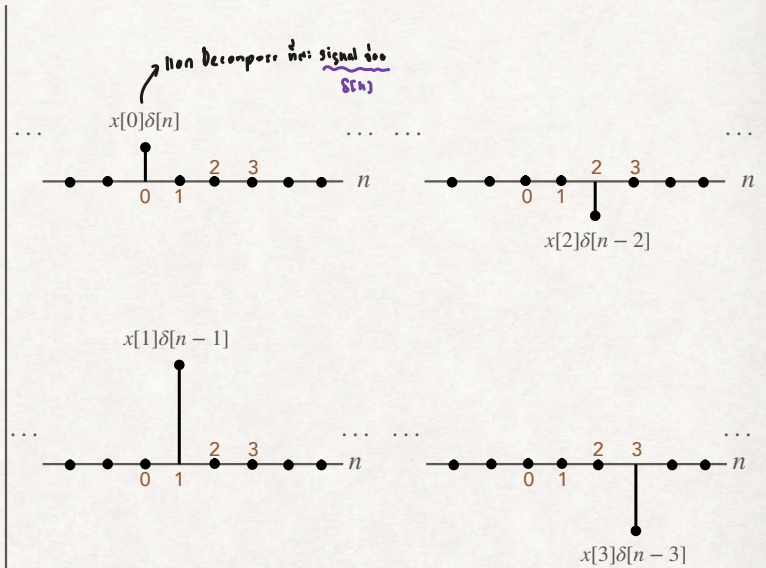
$$\text{Assume } m = n-k \rightarrow y[n] = \sum_{m=-\infty}^{\infty} x[m-n] h[m]$$

DISCRETE-TIME SIGNAL CONVOLUTION

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$



$$x[n] = x[0]\delta[n] + x[1]\delta[n-1] + x[2]\delta[n-2] + x[3]\delta[n-3]$$



Decomposition of a discrete-time signal

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GRAPHICAL INTERPRETATION OF CONVOLUTION

CAH $\Rightarrow$ Filter Pattern

EXERCISE 2.1



Find the convolution $y[n] = h[n] * x[n]$ of the following signals:

$$h[n] = \{1, 2, 0, -1, 1\}, \quad x[n] = \{1, 3, -1, -2\}$$

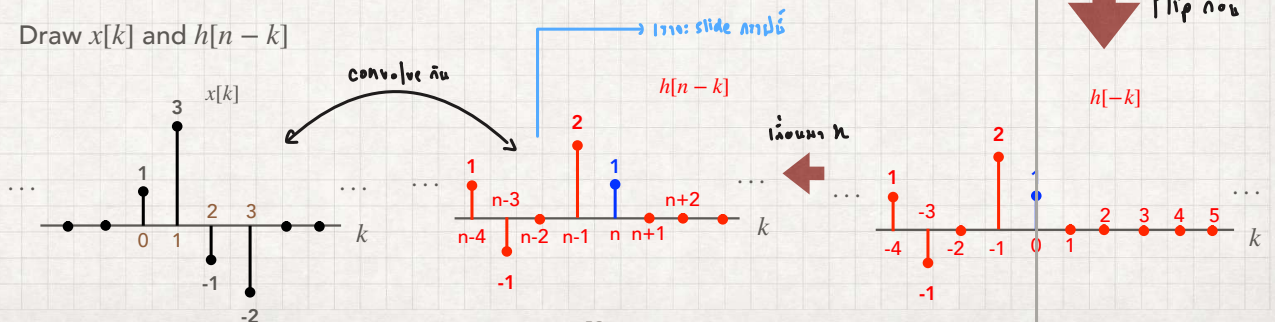
Partial Convolution: $y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$

Ans. if M is the length of $h[n]$ and N is the length of $x[n]$

total points = $M + N - 1$ starting from minimum index between $h[n]$ and $x[n]$

In this case, total points = $5 + 4 - 1 = 8$ points starting from 0 to 7

Draw $x[k]$ and $h[n-k]$



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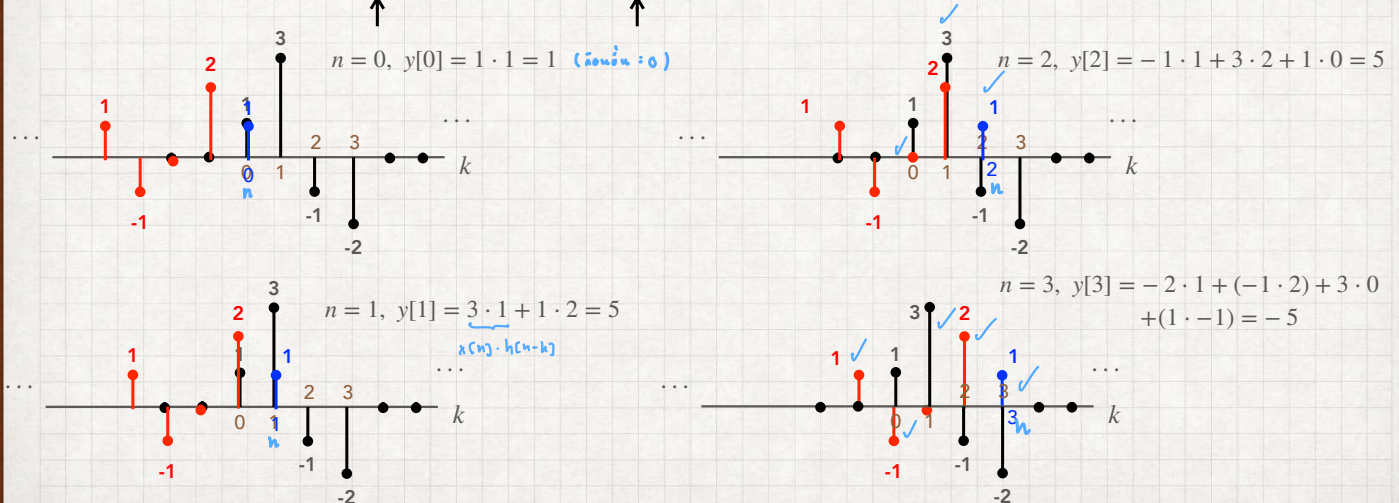
GRAPHICAL INTERPRETATION OF CONVOLUTION

EXERCISE 2.1



Find the convolution $y[n] = h[n] * x[n]$ of the following signals:

$$h[n] = \{1, 2, 0, -1, 1\}, \quad x[n] = \{1, 3, -1, -2\}$$



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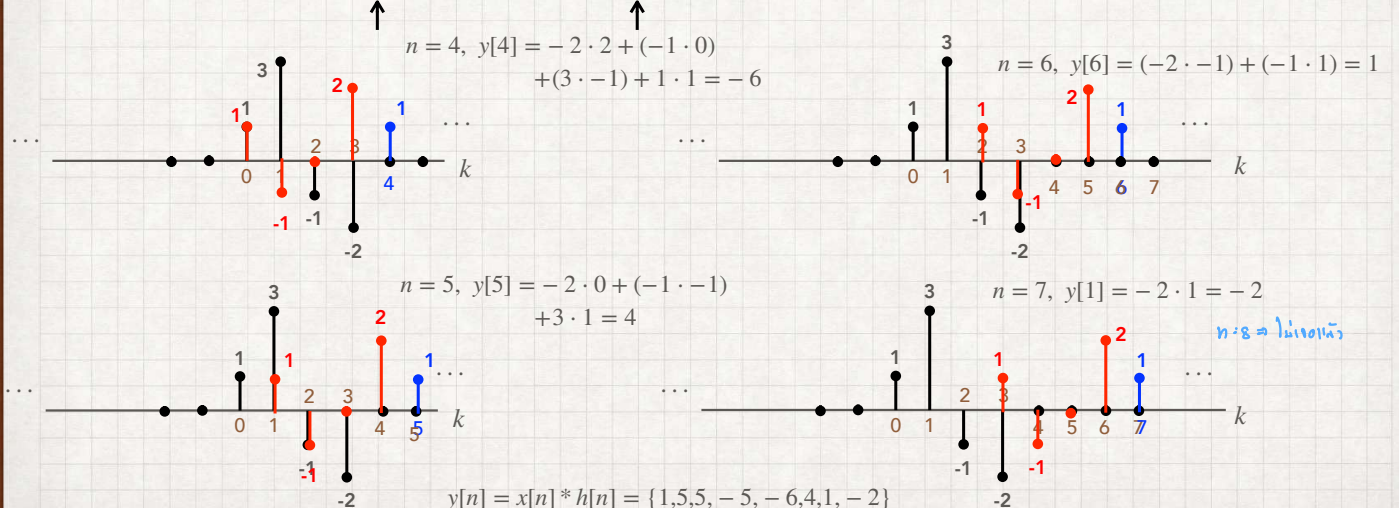
GRAPHICAL INTERPRETATION OF CONVOLUTION

EXERCISE 2.1



Find the convolution $y[n] = h[n] * x[n]$ of the following signals:

$$h[n] = \{1, 2, 0, -1, 1\}, \quad x[n] = \{1, 3, -1, -2\}$$



$$y[n] = x[n] * h[n] = \{1, 5, 5, -5, -6, 4, 1, -2\}$$

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Note: if $n < 0$ or $n > 7$

$$\begin{array}{r}
 \leftarrow 1 \ 2 \ 0 \ -1 \ 1 \leftarrow x[k] \\
 1 \ 3 \ -1 \ -2 \leftarrow h[n-k] \\
 \hline
 1 \ 2 \ 0 \ -1 \ 1 \\
 3 \ 6 \ 0 \ -3 \ 3 \\
 -1 \ -2 \ 0 \ 1 \ -1 \\
 -2 \ -4 \ 0 \ 2 \ -2 \\
 \hline
 1 \ 5 \ 5 \ -5 \ -6 \ 4 \ 1 \ -2
 \end{array}$$

GRAPHICAL INTERPRETATION OF CONVOLUTION

EXERCISE 2.1



Find the convolution $y[n] = h[n] * x[n]$ of the following signals:

$$h[n] = \{1, 2, 0, -1, 1\}, \quad x[n] = \{1, 3, -1, -2\}$$

Tabular method

	k	0	1	2	3	
		1	3	-1	-2	$\leftarrow x[n]$
						$\leftarrow$ Reversed $h[n]$
$y[0]$	1	-1	0	2	1	= 1
$y[1]$		1	-1	0	2	= 2+3 = 5
$y[2]$			1	-1	0	= 0+6-1 = 5
$y[3]$				1	-1	= -1+0-2-2 = -5
$y[4]$					1	= 1-3+0-4 = -6
$y[5]$						= 3+1+0 = 4
$y[6]$						= -1+2 = 1
$y[7]$						= -2

$$y[n] = x[n] * h[n] = \{1, 5, 5, -5, -6, 4, 1, -2\}$$

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GRAPHICAL INTERPRETATION OF CONVOLUTION

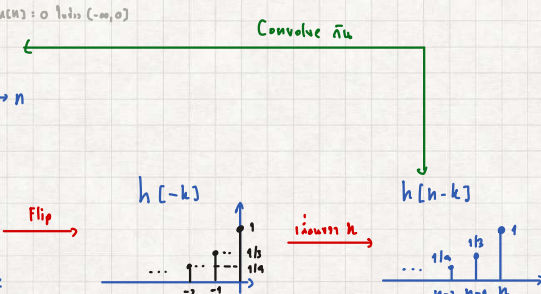
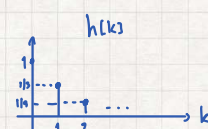
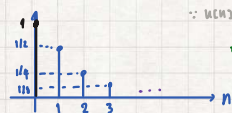
EXERCISE 2.2



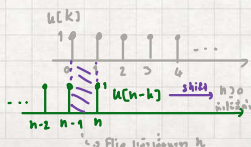
Let input $x[n] = \alpha^n u[n]$ and impulse response $h[n] = \beta^n u[n]$. Find an output, $y[n]$

$$x[n] = \left(\frac{1}{2}\right)^n u[n]$$

$$h[n] = \beta^n u[n]$$



Note



$$u[k] \cdot u[n-k] = \begin{cases} 1 & ; k \geq 0, n-k \geq 0 \\ 0 & ; \text{otherwise} \end{cases}$$

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} x[k] h[n-k] = \sum_{k=0}^n \left(\frac{1}{2}\right)^k \beta^{n-k} \\ &= \beta^n \sum_{k=0}^n \left(\frac{1}{2\beta}\right)^k \\ &= \beta^n \left(1 + \left(\frac{1}{2\beta}\right) + \left(\frac{1}{2\beta}\right)^2 + \dots + \left(\frac{1}{2\beta}\right)^n\right) \end{aligned}$$

Note: Geometric Series $\sum_{k=0}^n a r^k = a \frac{1-r^{n+1}}{1-r}$

Ans: $y[n] = \frac{\beta^n (1 - (\frac{1}{2\beta})^{n+1})}{1 - \frac{1}{2\beta}}$

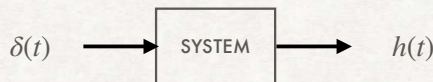
$$\text{Ans. } y[n] = \frac{\beta^n (1 - (\frac{\alpha}{\beta})^{n+1})}{1 - \frac{\alpha}{\beta}} u[n]$$

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CONTINUOUS-TIME SIGNAL CONVOLUTION



Given $h(t)$ is the response of a linear system to the input $\delta(t)$. *Impulse Response*



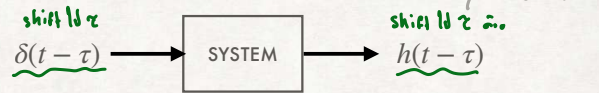
For the continuous-time system, input $x(t)$ can be written as the weighted sum of shifted unite-sampled functions:

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

integrate in tau domain

Discrete ; $x[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot \delta[n-k]$

If the system is time-invariance,



From superposition properties, (Linearity)

$$y(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau \rightarrow \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau = x(t) * h(t)$$

Discrete

"CONVOLUTION"

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CONVOLUTION PROPERTIES

CT convolution satisfies the following important properties

- Commutativity: $x(t) * h(t) = h(t) * x(t)$ *commutative*
- Associativity : $x(t) * h_1(t) * h_2(t) = [x(t) * h_1(t)] * h_2(t) = x(t) * [h_1(t) * h_2(t)]$ *no convolution is*
- Distributivity : $x(t) * [h_1(t) + h_2(t)] = [x(t) * h_1(t)] + [x(t) * h_2(t)]$

An LTI system with impulse response $h(t) = \delta(t)$ is the identity system

$$x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau) u(t - \tau) d\tau = \int_{-\infty}^t x(\tau) d\tau$$

Also applied for discrete-time convolution.

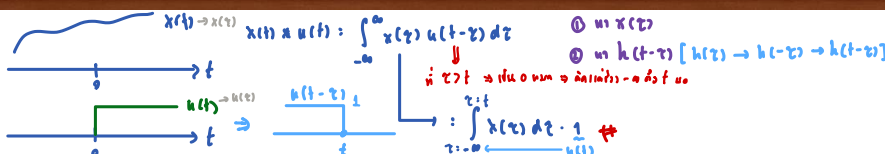
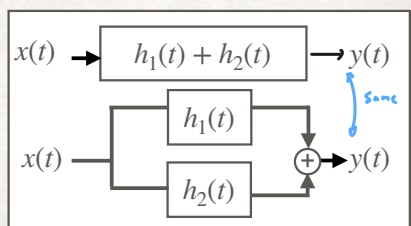
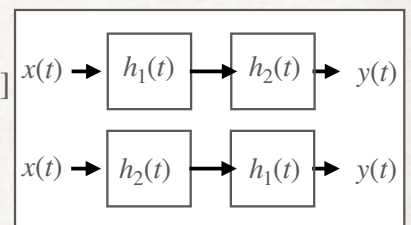
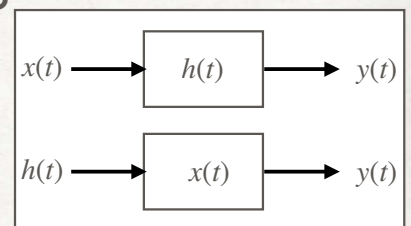


$x(t) * \delta(t) = x(t)$

$\delta(t) * x(t) = x(t)$

$\delta(t) * \delta(t) = \delta(t)$ [Identity function]

Step from Convolution



Note $\delta(t) * \delta(t) = \delta(t)$

CONVOLUTION EXAMPLE

EXERCISE 2.3



Consider the LTI system with impulse response $h(t) = e^{-at}u(t)$, $a > 0$

Input to the system is $x(t) = e^{-bt}u(t)$, $b \neq a$, $b > 0$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = \int_{-\infty}^{\infty} e^{-b\tau} u(\tau) e^{-a(t-\tau)} u(t-\tau) d\tau$$

Notes: $e^{-at} \cdot e^{a\tau}$ Chain Rule

$$= e^{-at} \int_{-\infty}^{\infty} e^{-(b-a)\tau} u(\tau) u(t-\tau) d\tau$$



for $t \geq 0$ is t
 for $t < 0$ is 0

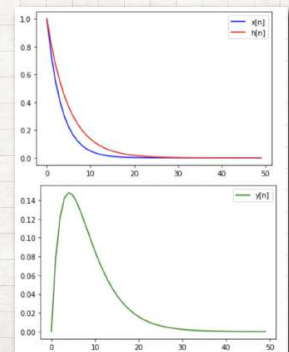
$$= e^{-at} \int_0^t e^{-(b-a)\tau} d\tau; t \geq 0$$

$$= e^{-at} \left[\frac{e^{-(b-a)\tau}}{-(b-a)} \right]_{\tau=0}^t; t \geq 0$$

$$= \frac{e^{-at}}{-(b-a)} [e^{-(b-a)t} - 1] u(t) = \frac{e^{-bt} - e^{-at}}{a-b} u(t)$$

$$y(t) = \frac{1}{a-b} (e^{-bt} - e^{-at}) u(t)$$

If $a = 3$, $b = 2$,

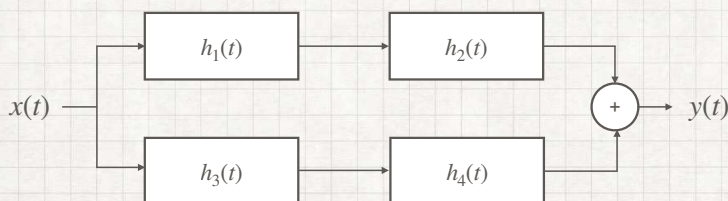


CONVOLUTION EXAMPLE

EXERCISE 2.4



Find the impulse response of the system shown in the figure below if



$$h_1(t) = e^{-2t}u(t)$$

$$h_2(t) = 2e^{-t}u(t)$$

$$h_3(t) = e^{-3t}u(t)$$

$$h_4(t) = 4\delta(t)$$

GRAPHICAL CONVOLUTION INTEGRATION

- When two signals are continuous for all t , convolution is just an ordinary integration. However, when one/both of the signals defined a piecewise fashion, we can use a graphical interpretation of convolution.

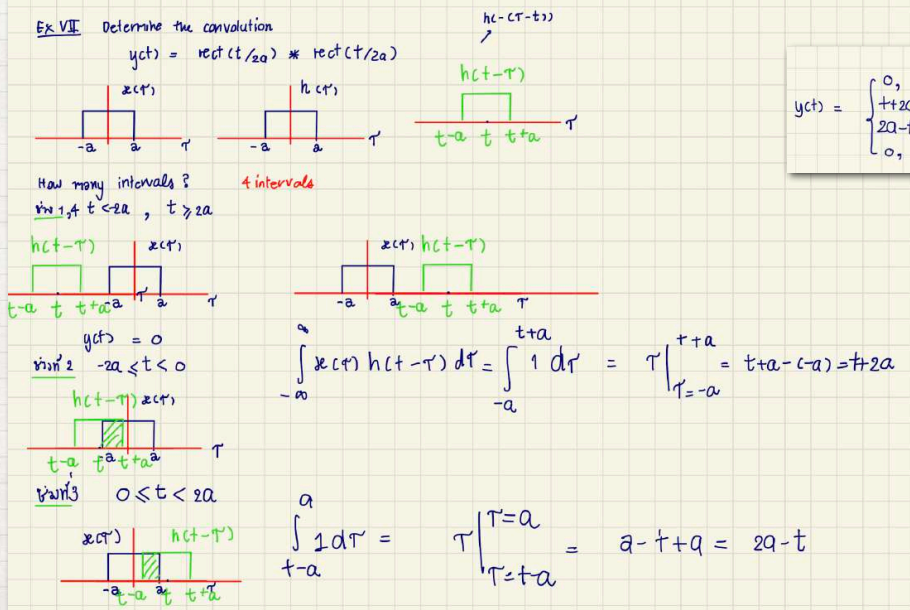


- To compute graphical convolution integration in the interval $t_{i-1} \leq t \leq t_i$
 - Step 1:** fixed value of t in $[t_{i-1}, t_i]$ and **plot** $x(t)$ and $h(t - \tau)$ and **the product** $g(t, \tau) = x(\tau) h(t - \tau)$ as a function of τ .
 - Step 2:** **integrate product** $g(t, \tau)$ as a function of τ (the integration is the area under the curve). Note that the integral $g(t, \tau)$ depends on t and τ . But τ **will disappear** when the integration is completed.

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GRAPHICAL INTERPRETATION OF CONVOLUTION

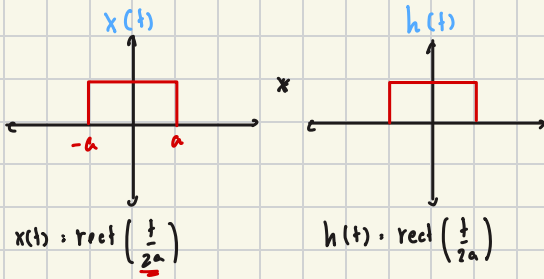
EXERCISE 2.5



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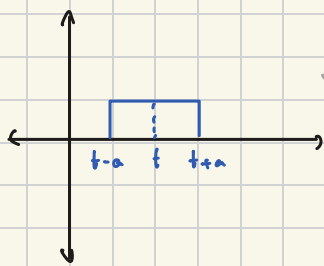
Ex: Given rect $2a \Rightarrow$ is convolution? $\Rightarrow$ Graph $y(t)$

$$y(t) = \int_{-\infty}^{\infty} \underbrace{x(\tau)}_{\text{fixed}} \underbrace{h(t-\tau)}_{\text{moving}} d\tau$$

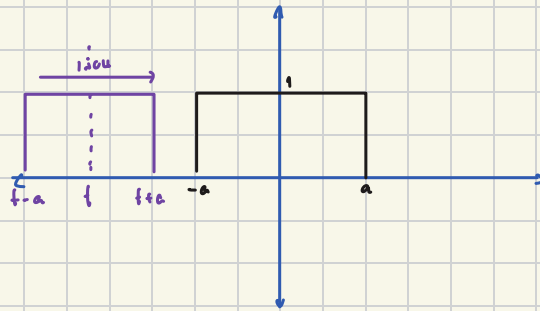


Q: Ođigōn
 $\therefore$ Nōiōn: a

$h(\tau) \rightarrow h(-\tau) \rightarrow h(t-\tau)$
 shift left
 lāiōn
 $\therefore$ Symmetry

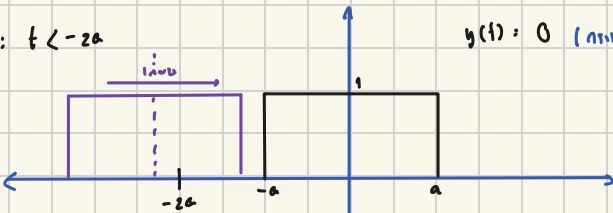


positive Shift

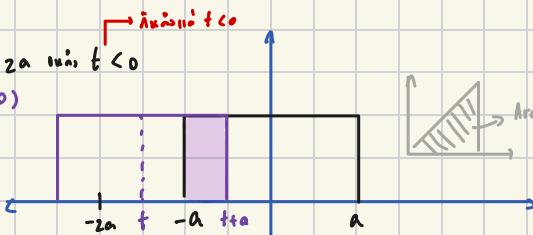


Case 1: $t < -2a$

$y(t) = 0$ (no overlap)



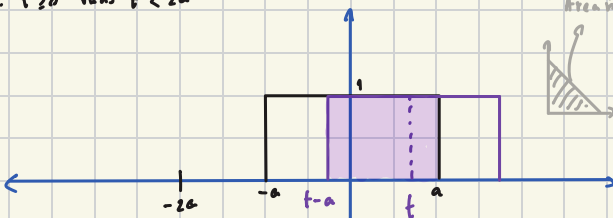
Case 2: $t \geq -2a$ and $t < 0$
 $[-2a, 0)$



Area: $\frac{1}{2} \times \text{base} \times \text{height}$
 Area: $\frac{1}{2} \times (t + 2a) \times 1$

$$y(t) = \int_{-a}^{t+a} h(t-\tau) d\tau = \left[\tau \right]_{-a}^{t+a} = t+a - (-a) = t+2a$$

Case 3: $t \geq 0$ and $t < 2a$



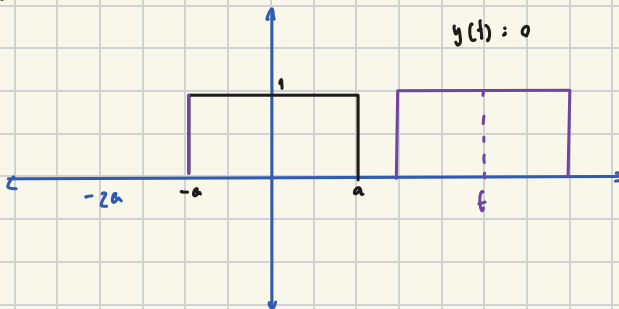
Area: $\frac{1}{2} \times \text{base} \times \text{height}$



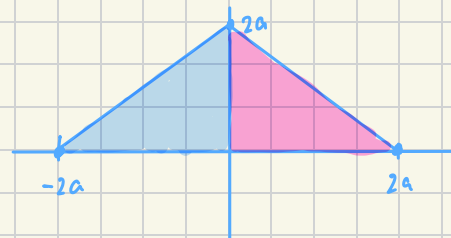
$$y(t) = \int_{t-a}^a h(t-\tau) d\tau = \left[\tau \right]_{t-a}^a = a - (t-a) = a - t + a = 2a - t$$

Case 4: $t \geq 2a$

$y(t) = 0$



$$\therefore y(t) = \begin{cases} 0 & t < -2a \\ t+2a & -2a \leq t < 0 \\ -t+2a & 0 \leq t < 2a \\ 0 & t \geq 2a \end{cases} \rightarrow \begin{cases} 0 & |t| \leq 2a \\ -|t| + 2a & \text{otherwise} \end{cases}$$



GRAPHICAL INTERPRETATION OF CONVOLUTION

EXERCISE 2.6



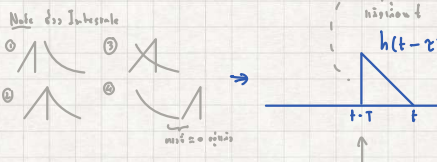
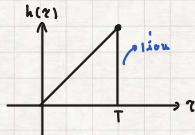
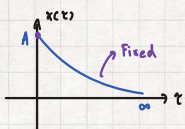
Consider the signals where

$$x(t) = Ae^{-t}, \quad 0 \leq t < \infty$$

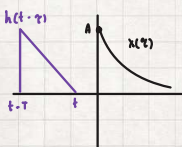
$$h(t) = \frac{t}{T}, \quad 0 \leq t < T$$

Find convolution using the graphical interpretation

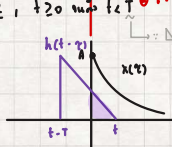
Plot $x(\tau)$, $h(\tau)$



Case 1: $t < 0 \Rightarrow y(t) = 0$

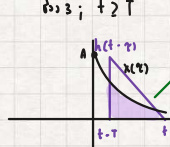


Case 2: $0 \leq t < T$



$$y(t) = \int_0^t Ae^{-\tau} \cdot (t-\tau) d\tau$$

Case 3: $t \geq T$



$$y(t) = \int_{t-T}^t Ae^{-\tau} \cdot (t-\tau) d\tau$$

$$y(t) = \frac{A}{T} (e^t + t - 1)$$

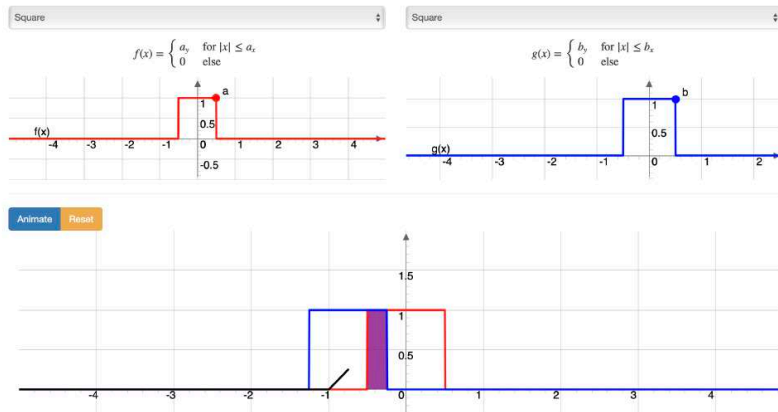
$$y(t) = \frac{A}{T} e^{-(t-T)} \cdot (e^{-T} + t - 1)$$

$$\begin{aligned} y_2(t) &= \int_0^t Ae^{-\tau} \cdot \frac{(t-\tau)}{T} d\tau \\ &= \frac{A}{T} \int_0^t e^{-\tau} (t-\tau) d\tau \\ &= \frac{A}{T} \left(t \int_0^t e^{-\tau} d\tau - \int_0^t \tau e^{-\tau} d\tau \right) \\ &= \frac{A}{T} \left(t [-e^{-\tau}]_0^t - \left[-\tau e^{-\tau} - \int -e^{-\tau} d\tau \right]_0^t \right) \\ &= \frac{A}{T} \left(-te^{-t} + t - \left[-te^{-t} - (-e^{-t}) \right] \right) \\ &= \frac{A}{T} (-te^{-t} + t + te^{-t} - e^{-t}) \\ &= \frac{A}{T} (t - e^{-t}) \end{aligned}$$

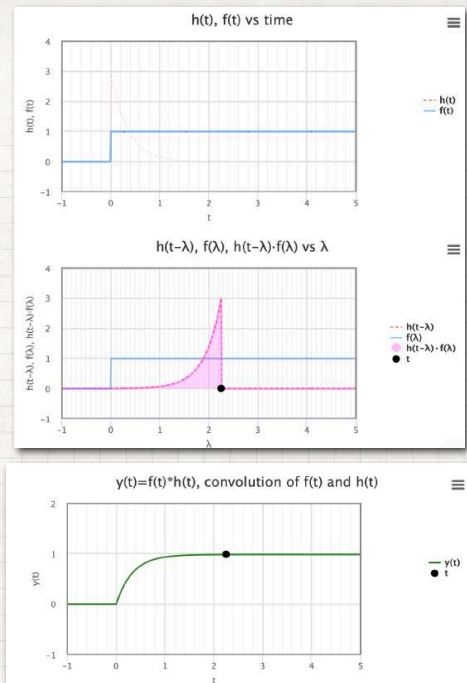
$$\begin{aligned} y_3(t) &= \int_{t-T}^t Ae^{-\tau} \cdot \frac{(t-\tau)}{T} d\tau \\ &= \frac{A}{T} \int_{t-T}^t e^{-\tau} (t-\tau) d\tau \\ &= \frac{A}{T} \left(t \int_{t-T}^t e^{-\tau} d\tau - \int_{t-T}^t \tau e^{-\tau} d\tau \right) \\ &= \frac{A}{T} \left(t [-e^{-\tau}]_{t-T}^t - \left[-\tau e^{-\tau} - \int -e^{-\tau} d\tau \right]_{t-T}^t \right) \\ &= \frac{A}{T} \left(t(-e^{-t} + e^{-(t-T)}) - \left[-te^{-t} - (-e^{-t}) - (-\tau e^{-\tau} - (-e^{-\tau})) \right]_{t-T}^t \right) \\ &= \frac{A}{T} \left(-te^{-t} + te^{-(t-T)} - \left[-te^{-t} - (-e^{-t}) + \tau e^{-\tau} + e^{-\tau} \right]_{t-T}^t \right) \\ &= \frac{A}{T} \left(-te^{-t} + te^{-(t-T)} + te^{-t} + e^{-t} - \tau e^{-\tau} - e^{-\tau} \right)_{t-T}^t \\ &= \frac{A}{T} \left(e^{-t} + e^{-(t-T)} - e^{-(t-T)} \right) = \frac{A}{T} e^{-(t-T)} (e^{-T} + t - 1) \end{aligned}$$

DEMO

Convolution demo



<https://phiresky.github.io/convolution-demo/>



<https://lpsa.swarthmore.edu/Convolution/CI.html>

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- Alan V. Oppenheim, Alan S. Willsky and S. Hamid Nawab., **Signals and Systems**, 2nd Edition, Prentice Hall (1996)
 - Chapter 2 LTI systems - discrete-time and continuous-time convolution integrals